Replicating the Fama-French Factor Models on European Equities (1990-2026)

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Part I -- The academic questions behind the design

Every empirical asset-pricing model is an answer to a prior puzzle. This section organizes the questions that motivated the Fama-French factor design, and the knowledge needed to understand the answers.

Q1. Why do average returns differ across stocks?

The foundational question of asset pricing: in a market where investors are risk-averse, expected returns should compensate for risk. The Capital Asset Pricing Model (Sharpe, 1964; Lintner, 1965) gives the sharpest version of this idea -- a single number, the market beta, should explain all cross-sectional differences in expected returns.

$$E[R_i] - R_f = \beta_i * (E[R_m] - R_f)$$

If the CAPM held exactly, sorting stocks by anything other than beta -- size, valuation ratios, past returns -- would reveal no return differences. The entire factor literature exists because it does not.

Related knowledge: expected vs. realized returns (we only ever observe realized returns; a factor 'premium' is the historical average of realized returns, an estimate of the ex-ante risk premium, not a guarantee); the joint-hypothesis problem (any test of market efficiency is simultaneously a test of the asset-pricing model used -- Fama, 1970).

Q2. Is the CAPM enough? The size and value anomalies

- Size (Banz, 1981): small-capitalization stocks earned higher average returns than their market betas predicted.
- Value (Fama & French, 1992): high book-to-market stocks earned higher average returns than growth stocks, even after controlling for beta and size. Once size and book-to-market were accounted for, market beta had almost no explanatory power left.

Fama and French (1993) turned these anomalies into factors: tradable long-short portfolios -- SMB (small minus big) and HML (high minus low book-to-market) -- and showed that a three-factor model absorbed most of the CAPM's pricing errors.

Related knowledge: anomaly (a return pattern unexplained by the benchmark model); sorting (forming portfolios by ranking stocks on a characteristic); long-short factor construction (e.g., 2x3 independent sorts on size and book-to-market).

Q3. Risk or mispricing? Competing explanations of the premia

- Risk-based: value firms tend to be distressed; their high returns compensate investors for bearing distress/bankruptcy risk that market beta misses (Fama & French, 1996).
- Behavioral: investors over-extrapolate past growth, overpaying for glamour stocks; the premium is the slow correction of mispricing (Lakonishok, Shleifer & Vishny, 1994).

This debate is unresolved -- but it is why the literature keeps score with alphas: any return the factors cannot explain is either missing risk or mispricing, and the GRS test (Part IV) is the formal scoreboard.

Related knowledge: alpha (abnormal return vs. the model); limits of arbitrage (why mispricing can persist).

Q4. Do U.S. results travel? Why Europe is the right out-of-sample test

A pattern found on U.S. data may be data mining -- an accidental discovery from searching many specifications on one dataset. The scientific response is out-of-sample testing: the same sorts, the same regressions, on data the original authors did not use to design the model. Europe is the natural laboratory: deep, liquid markets, long histories, and economic structures different enough from the U.S. that survival there is informative.

Q5. From three factors to five: the valuation-equation logic

The five-factor model was not assembled by data mining. It follows from the Miller-Modigliani dividend-discount identity: holding everything else fixed, a stock's expected return is higher when its book-to-market ratio is higher, expected profitability is higher, and expected investment is lower. Profitability (Novy-Marx, 2013) and investment (Aharoni, Grundy & Zeng, 2013) patterns in returns therefore had to be related to expected returns -- the only question was whether the market already priced them. Fama and French (2015) added RMW and CMA and showed the five-factor model absorbed the anomalies -- notably the low returns of small, unprofitable, high-investment firms -- that embarrassed the three-factor model.

Q6. The factor zoo: why replication discipline matters

Hundreds of 'factors' have been proposed (Cochrane, 2011, calls it the factor zoo). Most do not survive transaction costs, out-of-sample tests, or multiple-testing corrections. This replication practices the discipline the literature demands: pre-committed test assets (the 25 Size-B/M portfolios), a fixed sample, and a formal joint test (GRS) rather than cherry-picked t-statistics.

Part II -- Literature (concise tour)

Study	Contribution
Sharpe (1964); Lintner (1965)	CAPM: market beta as the single priced risk
Banz (1981)	Size effect
Fama & French (1992)	B/M and size drive the cross-section; beta fades
Fama & French (1993)	Three-factor model: Mkt-RF, SMB, HML
Lakonishok et al. (1994)	Behavioral account of the value premium
Fama & French (1996)	Risk-based multifactor explanation of anomalies
Carhart (1997)	Momentum (WML) as a fourth factor
Novy-Marx (2013)	Gross profitability premium -> motivates RMW
Aharoni et al. (2013)	Investment effect -> motivates CMA
Fama & French (2012)	Size, value, momentum in international returns
Fama & French (2015)	Five-factor model adds RMW and CMA
Gibbons, Ross & Shanken (1989)	GRS F-test of joint alpha = 0
Cochrane (2011)	'Factor zoo' critique; discount-rate view

Full citations: REFERENCES.md.

Part III -- Data

- Source: Kenneth R. French Data Library, files constructed from the August 2026 Bloomberg database (free for research use; vendored in data/raw/).
- Region: Europe (16 developed markets incl. the UK, Germany, France, Switzerland).
- Factors (monthly): Mkt-RF, SMB, HML, RMW, CMA, RF. Factors are returns on diversified long-short portfolios from 2x3 sorts.
- Test assets (monthly): 25 value-weighted portfolios from independent 5x5 sorts on market

capitalization (Size) and book-to-market -- the classic Fama-French (1993) test assets, reformed each June.

- Sample: July 1990 - August 2026, T = 434 months. Missing values (-99.99) treated as NaN; the common sample is used throughout.

Part IV -- Methodology

For each test portfolio i, estimate by OLS:

*R(i,t) - Rf(t) = alpha(i) + b(i)*MKT(t) + s(i)*SMB(t) + h(i)*HML(t) [+ r(i)*RMW(t) + c(i)*CMA(t)] + e(i,t)*

Reported per portfolio: annualized alpha, t(alpha), factor loadings with t-statistics, adjusted R-squared. Model comparison uses (1) the count of significant alphas (|t| > 2) out of 25, (2) mean adjusted R-squared, and (3) the Gibbons-Ross-Shanken F-test of H0: all 25 alphas = 0.

*F = (T/N) * [(T-N-K)/(T-K-1)] * (a' S^-1 a) / (1 + mu_f' Omega^-1 mu_f), distributed F(N, T-N-K),*

where N = 25 test assets, K factors, S the residual covariance matrix, and mu_f, Omega the factor mean vector and covariance matrix. Rejection means the model's factors cannot jointly price the test assets.

Note on standard errors: t-statistics use OLS standard errors. Returns show mild autocorrelation/heteroskedasticity; Newey-West corrections would be a natural robustness extension (see Part VI).

Part V -- Results

5.1 Factor premia in Europe, 1990-2026

Factor	Ann. mean	Ann. vol	t(mean)	Sharpe
Mkt-RF	6.36%	17.01%	2.25	0.37
SMB	0.15%	7.23%	0.12	0.02
HML	4.24%	9.01%	2.83	0.47
RMW	3.60%	5.56%	3.89	0.65
CMA	1.71%	6.21%	1.66	0.28

Three facts stand out. First, the size premium is economically and statistically zero in Europe (0.15%/yr, t = 0.12) -- consistent with the documented post-publication decay of the size effect. Second, value and profitability premia are alive: HML earns 4.24%/yr (t = 2.83) and RMW 3.60%/yr (t = 3.89, Sharpe 0.65, the strongest risk-adjusted premium). Third, the investment premium is positive but marginal (t = 1.66).

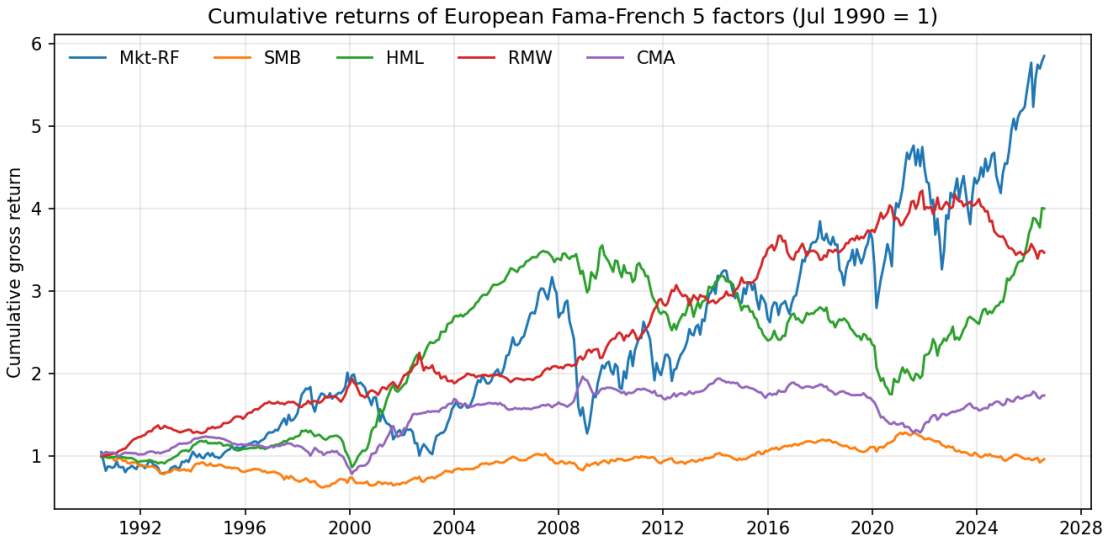


Figure 1. Cumulative returns of the five European Fama-French factors (July 1990 = 1).

5.2 Do the models price the 25 Size-B/M portfolios?

- Factor loadings behave as theory predicts. Within every size quintile, HML loadings rise monotonically with book-to-market (smallest quintile: -0.43, -0.24, -0.10, 0.09, 0.35). Mean adjusted R-squared is 0.942 (FF3) and 0.946 (FF5).

- The small-growth problem replicates -- and the five-factor model halves it. The small-growth portfolio earns an FF3 alpha of -5.16%/yr (t = -4.94), the largest pricing error in the cross-section. Under FF5 it shrinks to -2.43%/yr (t = -2.38): negative RMW (-0.48) and CMA (-0.26) loadings show these firms are unprofitable high-investors -- exactly the anomaly the two added factors were designed to absorb. Big-value's FF3 alpha (-2.34%/yr, t = -2.44) likewise vanishes under FF5 (+0.06%/yr).

- Scoreboard: significant alphas (|t| > 2) fall from 8 of 25 (FF3) to 6 of 25 (FF5). The GRS test rejects joint alpha = 0 for both models -- FF3: F = 3.05 (p < 0.001); FF5: F = 2.71 (p < 0.001) -- so neither model fully prices European equities, but the five-factor model prices them better, mirroring the U.S. verdict of Fama and French (2015).

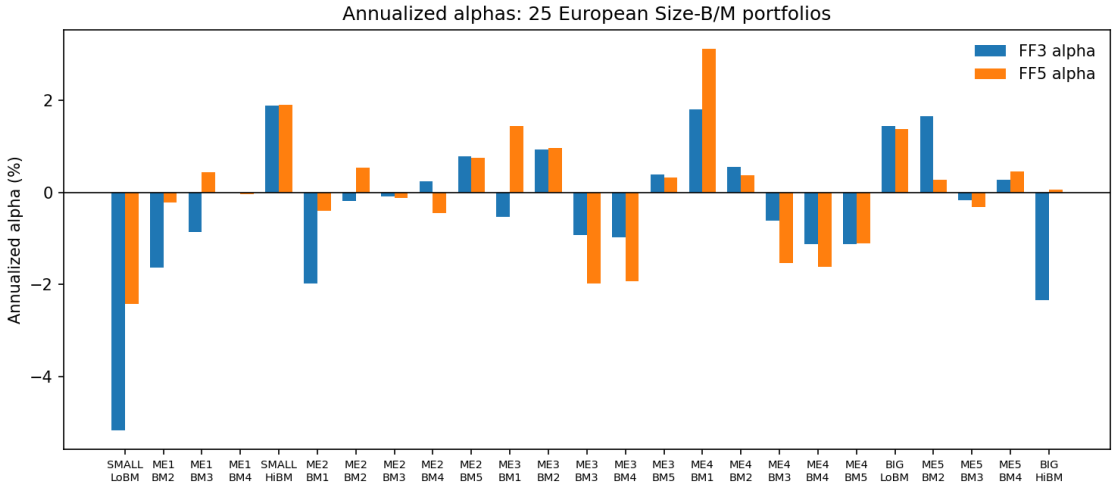


Figure 2. Annualized alphas: 25 European Size-B/M portfolios, FF3 vs FF5.

5.3 Dynamics

HML's 12-month rolling mean shows the value premium is highly time-varying: deeply negative during the late-1990s tech boom and the 2020 growth rally (HML lost -30.1% in 2020 alone), violently positive in recoveries (+29.6% in 2025). The premium is compensation for enduring bad times, not a free lunch -- the core intuition of the risk-based story in Q3.



Figure 3. HML (value premium), 12-month rolling mean, Europe.

Part VI -- Conclusion and limitations

What we learn

- The value and profitability premia survive out-of-sample in Europe; the size premium does not.
- The five-factor model's added factors earn their keep precisely where theory said they would -- small, unprofitable, high-investment firms.
- Formally, both models are rejected: factor models are useful approximations, not literal truth.

Limitations

- Test assets are value-weighted portfolios, tradable only approximately; no transaction costs are modeled.
- OLS standard errors -- Newey-West (1987) corrections are a standard robustness check, omitted here for clarity.
- One region, one sort. Results are conditional on the 25 Size-B/M design; other test assets could tell a different story -- the classic joint-hypothesis caveat.
- No investment implication. Historical premia do not predict future returns; see the disclaimer in the README.

References & glossary

Full citations: REFERENCES.md. Terminology (English + Chinese): GLOSSARY.md.